

Prerequisites for testing

The prerequisites that need to be taken care of are:

Setting up the IoT hub. An IoT hub is a server that can talk to IoT devices and collect data from them. It may be a Web server on cloud, or an app on a mobile device.

Setting up the IoT device. It is essential to make sure that the IoT device is turned on, accessible and creates a genuine use case for testing.

Setting up the network. The device(s) and hub can be connected via Wi-Fi, Bluetooth, satellite signals, NFC, cellular, Zigbee or LoRa. Each technology has its pros and cons. It can be a challenge to decide which connectivity protocol to use and then to figure out how to test to make sure the rollout goes smoothly.

Design and simulation tools

There are various tools used during testing IoT systems. Greater integration delivers lower cost, lower energy consumption and better performance. It is important to know how to choose a solution that can solve the challenges, and to know the best measurement tips and practices that need to be followed.

Software. An open source application called Wireshark can be used for monitoring traffic in the interface, source/destination host addresses and more. It helps users in displaying TCP/IP and other packets that are transmitted or received over a network, as in Tcpdump.

Hardware. This helps in debugging the target platform code and shows variable step-by-step procedure, for example, JTAG dongle. It can be used to check various events with time stamps, glitches in power supply, signal integrity using a digital storage oscilloscope and so on.

A software-defined radio is used to emulate the receiver and transmitter for a large range of wireless gateways.

Architecture for the IoT comprises multiple layers built on top of each other. To create industry-specific solutions, components in each layer need to work in sync, to effectively convert data into information.

Latest IIoT testing and monitoring devices

Some testing and monitoring devices for the IIoT are:

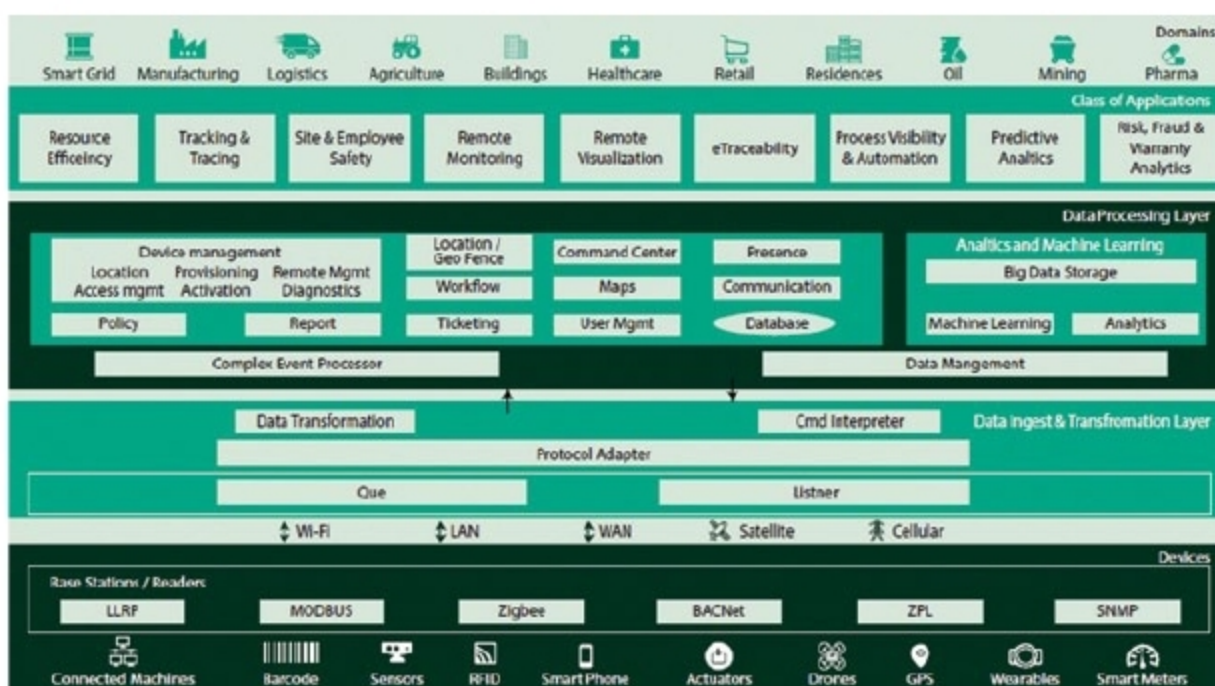


Fig. 2: The IIoT is a web of interconnected layers (Credit: www.infosys.com)

- imc's CRONOScompact provides adaptable measurement and control system for mixed-signal testing.

- Analog Devices' AD8233 electrocardiogram integrated AFE has microamp-range power requirements that enable extended battery life and continuous monitoring.

- Texas Instruments' industrial monitor with Sitara processor addresses the increasing demand for industrial communication connectivity and processor performance with superior video processing and graphic performance.

Challenges for IoT testers

Common challenges that IoT testers face are calibration of sensors that can vary from device to device, setting up the environment, connectivity, power, security and privacy, hardware quality and accuracy, among others.

Hardware-software mesh. The IoT is an architecture comprising hardware, software, sensors and communication gateways. And testing in terms of environment, data transfer and the like is tedious.

Device interaction module. When hardware and software integrate with each other, things such as security, backward compatibility and upgrade issues become a challenge for the testing team, as related sub-systems and components can be owned by third-party units.

User interface. Testing devices for every platform—iOS, Android, Windows and Linux, among others—is a challenge.

Network connection. An IoT archi-

ture must be tested for all kinds of network connectivity speeds. For this, virtual network simulators are used to vary network load, connectivity, stability and so on. Real-time networks can be challenging in the long run.

Dynamic environment. The IoT has a dynamic environment with millions of sensors and devices in conjunction with intelligent software. A complex set of use cases must be created to test cases and data.

Real-time complexity. IoT applications can have multiple real-time scenarios, and use cases are extremely complex. Getting regulatory check points or the system deployed is tough.

To summarise

Anisha Nikash Dumbre, research analyst, Frost & Sullivan, says, "By 2020, a number of IoT-enabled test systems are expected to be launched, catering to diverse services and fields within the industrial ecosystem. Democratisation of the IIoT will lead to cheaper software-as-a-service (SaaS) models."

Combining new technology with rapidly-developed software on newly-created hardware—where critical bugs concerning reliability, safety and performance are often overlooked—makes testing of IIoT devices difficult. Emergence of smart testing for the IIoT will have a significant impact on the test and measurement space. Advancements in technologies are making industries more effective and enabling machines to work smartly. Predictive analysis will make test systems intelligent, and their use will maximise digital connectivity. **EFY**